PULLBACK OF A STRICT NORMAL CROSSINGS DIVISOR BY A SMOOTH MORPHISM IS A STRICT NORMAL CROSSINGS DIVISOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file etale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 02OO: sources/stacks-chow/tags.txt, source lines 3523–3523. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 0BI9: sources/stacks-chow/tags.txt, source lines 14482–14482. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00TT: sources/stacks-chow/tags.txt, source lines 1087–1087. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00ON: sources/stacks-chow/tags.txt, source lines 901–901. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let $D \subset X$ be a strict normal crossings divisor. If $f : Y \to X$ is a smooth morphism of schemes, then the pullback f^*D is a strict normal crossings divisor on Y.*

Proof. As f is flat the pullback is defined by Divisors, Lemma 02OO hence the statement makes sense. Let $q \in f^*D$ map to $p \in D$. Choose a regular system of parameters $x_1, \ldots, x_d \in \mathfrak{m}_p$ and $1 \leq r \leq d$ as in Definition 0BI9. Since f is smooth the local ring homomorphism $\mathcal{O}_{X,p} \to \mathcal{O}_{Y,q}$ is flat and the fibre ring

$$\mathcal{O}_{Y,q}/\mathfrak{m}_p\mathcal{O}_{Y,q} = \mathcal{O}_{Y_p,q}$$

is a regular local ring (see for example Algebra, Lemma 00TT). Pick $y_1, \ldots, y_n \in \mathfrak{m}_q$ which map to a regular system of parameters in $\mathcal{O}_{Y_p,q}$. Then $x_1, \ldots, x_d, y_1, \ldots, y_n$ generate the maximal ideal $\mathfrak{m}_q$. Hence $\mathcal{O}_{Y,q}$ is a regular local ring of dimension $d + n$ by Algebra, Lemma 00ON and $x_1, \ldots, x_d, y_1, \ldots, y_n$ is a regular system of parameters. Since f^*D is cut out by $x_1 \ldots x_r$ in $\mathcal{O}_{Y,q}$ we conclude that the lemma is true. □

Date: stacks-project:etale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.